



A two-stage model to forecast elections in new democracies[☆]

Kenneth Bunker

University of Milan, Department of Social and Political Sciences, Via Conservatorio 7, Milan MI, 20122, Italy



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ABSTRACT

The purpose of this article is to propose a method to minimize the difference between electoral predictions and electoral results. It builds on findings that stem from established democracies, where most of the research has been carried out, but it focuses on filling the gap for developing nations, which have thus far been neglected by the literature. It proposes a two-stage model in which data are first collected, filtered and weighed according to biases, and then output using Bayesian algorithms and Markov chains. It tests the specification using data stemming from 11 Latin American countries. It shows that the model is remarkably accurate. In comparison to polls, not only does it produce more precise estimates for every election, but it also produces a more accurate forecast for nine out of every ten candidates. The article closes with a discussion on the limitations of the model and a proposal for future research.

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1. Introduction

The recent failure of pollsters to accurately predict a series of high-profile elections across the world has raised doubt to whether there is a fundamental problem in tracking public opinion sentiment when it comes to electoral cycles (Cassino, 2016; Cowley, 2017; Tao & Reis-Smart, 2014; Zukin, 2015). While there is no evidence that accuracy levels have decayed over time, there is no proof that they have improved either (Prosser & Mellon, 2018). This is problematic, since research suggests that the information that voters are exposed to determines both electoral participation rates (Lassen, 2005; Palfrey & Poole, 1987) and voter utility levels (Ghirardato & Katz, 2006; Matsusaka, 1995).

The purpose of this article is to contribute to the production of accurate information during electoral cycles. It proposes a model based on poll aggregation and state space representation to track public opinion and generate electoral forecasts. While the use of such models are relatively recent (see Silver, 2008), their efficiency

to aggregate data over time within the restrictions of probability theory has consolidated them as an essential tool to track voter preferences (Walther, 2015). And, while they are still evolving, they have often been able to generate remarkably accurate forecasts (see Jackson, 2016). But, because they have been mainly developed for use in established democracies, the full extension of their predictive capabilities is unknown.

This article seeks to bridge that gap. It borrows from the evidence accumulated in established democracies to advance a model that can accurately track public opinion and forecast elections in new democracies. It builds from models that have been developed for use in North America and Europe to develop a specification that can fit the particular institutional frameworks and political patterns of developing democracies. It particularly focuses on the new democracies of Latin America, a region in which the majority of countries elect presidents as formateurs (in contrast to prime ministers), and present comparatively high electoral volatility and party fragmentation levels.

The contribution is twofold. On the one hand, by implementing the model in an entirely novel context, future research will have more tools to evaluate the overall strengths and weaknesses of the method as a whole. On the other hand, by testing the model in a framework that

[☆] Data and code for replication are available on the author's website: <http://kennethbunker.github.io>.

E-mail address: kabunker@gmail.com.

has been thus far neglected by the literature, students of developing democracies will have additional resources to explore the dynamics that take place at the intersection between public opinion and elections. Furthermore, by making the model specifications openly available, it provides other scholars with a benchmark to both toggle the settings to increase specific accuracy levels and develop its complexity to reduce overall error parameters.

The remainder of this article is structured as follows. The following section briefly discusses the need to provide voters with accurate information and summarizes the manners through which it has been attempted so far. The third section shows that most of the research on electoral forecasting stems from established democracies and argues for the need to expand the literature into new democracies. The fourth section describes the two-stage model and its properties. The fifth section describes the data that are used to test the model, and the sixth section shows how it fairs in comparison to poll predictions. The final sections close with a discussion on the limitations of the model and proposals for future research.

2. Previous research

Voters use their knowledge to inform their decisions (Markus & Converse, 1979). Those with more information are not only more likely to vote (Bartels, 1996; Feddersen & Pesendorfer, 1996, 1999; Lassen, 2005; Palfrey & Poole, 1987) but are also more likely to vote for the candidate who yields them with the highest total utility (Ghirardato & Katz, 2006; Matsusaka, 1995). Democracies that institutionally account for voter misinformation bias, and aim at curbing the resulting asymmetries, tend to produce elections with higher rates of citizen participation as well as yield voters with higher levels of post-election satisfaction (see Carpinì & Keeter, 1996; Milner, 2002).

In contrast, voters with little or inaccurate information do not only vote less but are also less likely to report having voted for the “right candidate” (Matsusaka, 1995). Uninformed voters can also contribute to the production of artificial, and potentially harmful outcomes for representative democracy (Fowler & Margolis, 2014; Winters & Weitz-Shapiro, 2013). Naturally, uninformed voters misinform the electoral process (Blais, Gidengil, Fournier, & Nevitte, 2009; Larcinese, 2007). By making inefficient assumptions on the distribution of electoral preferences, they replicate and echo preexisting information biases (Nadeau, Cloutier, & Guay, 1993). These distortions are visible in phenomena as diverse as the bandwagon and underdog effects (Simon, 1954).

Until now, pre-electoral surveys (polls) have played the predominant role in providing information to voters (Banarjee, 1992). As one of the oldest and most established methods of tracking public opinion and producing election night predictions, both voters and media have used them to determine the “true” state of electoral preferences. By looking at both individual polls, as well as the overall trend of polls, they have sought to determine levels of competition and margins of victory (Skalaban, 1988). However, polls are not merely informative. They also have a direct effect on elections. Research shows

that voters who are aware of polls vote differently than those who are not (Morton, Muller, Page, & Torgler, 2015; Sudman, 1986).

Unfortunately, their failure to accurately predict results in a series of high-profile elections across the world—such as the United Kingdom 2015 General election, the United States 2016 Presidential election and the Australian 2019 Federal election—has raised widespread doubt on their ability to properly represent the electorate (Cassino, 2016; Cowley, 2017; Tao & Reis-Smart, 2014; Zukin, 2015). In some cases, polls were so far off their marks that they ended up sparking major investigations; as in the case of the UK 2015 General election and the BPC report *The Inquiry into the Failure of the 2015 Pre-election Polls*, and the 2019 Australian Federal election and the AMSRO report *Inquiry into the Performance of the Opinion Polls*.

Luckily, polls are not the only manner of tracking public opinion and generating election night forecasts. Lewis-Beck (2005) identifies two alternative methods: statistical models and political stock markets. Lewis-Beck and Tien (2012) further discriminate among polls, pundits, models and markets. Along the same lines, Graefe, Armstrong, Jones, and Cuzán (2014) differentiate among polls, experts, quantitative models and electronic betting markets. Their differences and similarities have been well documented, as well as their relative performances (cf. Armstrong, Green, & Graefe, 2015; Graefe et al., 2014; Kou & Sobel, 2004; Leigh & Wolfers, 2006; Lewis-Beck & Rice, 1984; Rothschild, 2009, 2015; Williams & Reade, 2016).

However, like polls, the alternative methods also have run into trouble. For example, recent research shows that experts and political insiders are as likely as the average voter to fall to wishful thinking (Sjöberg, 2009) and information (Calvillo & Rutchick, 2014) biases. Similarly, quantitative models have lost ground to more complex data models and specifications (Kennedy, Wojcik, & Lazer, 2017). In turn, betting markets have been put under scrutiny because of their vulnerability to manipulation (Graefe, 2017). In sum, while most of these alternative methods continue to be useful and used to track public opinion during electoral cycles, they have nonetheless failed to consistently and accurately predict election results.

It is no surprise, then, that scholars have resorted to alternatives to increase accuracy levels. While some have suggested cross-method combinations (see Armstrong et al., 2015; Lewis-Beck & Dassonneville, 2015), others have proposed within-method aggregation (see Silver, 2008; Wang, 2015). It is among the latter line of research that research has seen some of the most important developments (Pasek, 2015). For example, by aggregating polls scholars have been able to significantly improve predictions, often forecasting elections to the decimal (see Jackson, 2016). And while it is still a burgeoning subfield, models based on poll aggregation have already been able to rival traditional methods (Graefe, 2017).

Poll aggregation is simply the combination of data stemming from pre-election surveys. It is grounded in likelihood theories and bracketing principles (Larrick & Soll, 2006). To exemplify, take an election in which two

polls with similar characteristics are fielded with the intention of predicting the vote difference between the top two candidates. It is likely that the average of the two polls will be a better estimator of the parameter of interest than any poll chosen at random. It may be the case that one of the polls is more accurate than the aggregate value, but the aggregate value will never be the least accurate of the three estimators. Theoretically, as the amount of data increases (e.g., pollsters, polls, candidates, valid votes), the level of error decreases.

While poll aggregation models do not go without limitations (see Graefe, Küchenhoff, Stierle, & Riedl, 2015), they do provide solutions to many of the pitfalls traditional methods have not been able to circumvent. In contrast to polls and betting markets, they are less vulnerable to late swings and outliers; they do not generally take potentially biased information at face value. In comparison to quantitative models, they are more versatile; they can be easily designed to combine data from alternative sources, and while they start from relatively simple specifications, they can be designed to fit complex contexts. Finally, in comparison to experts, they are more likely to tend toward the average preference; they are more effective in detecting latent trends.

3. Dynamic linear models

Many of the studies that have used poll aggregation methods have done so in combination with space state dynamic linear models (DLMs) (see West & Harrison, 1997). In the realm of forecasting, DLMs have sought to estimate latent trends of support for parties (or candidates) and extrapolate those trends into the future (Hurtado Bodell, 2016). While they do not go without problems (see Graefe et al., 2015), they have nonetheless been found to produce accurate estimators (see Jackson, 2016). Their contribution has lied not only in their effectiveness of combining data over time within the restrictions of probability theory, but also in their ability to perform real-time tracking of electoral support with continuous information (Walther, 2015).

One of the first DLMs was proposed by Jackman (2005). In his seminal article, he proposed a Bayesian model that only required polls only as input. Since then, most of the research has stemmed from the United States. Rigdon, Jacobson, Tam Cho, Sewell, and Rigdon (2009) proposed a Bayesian forecasting model to predict Electoral College outcomes. Lock and Gelman (2010) used Bayesian inference to integrate the various sources of data and estimate vote shares for the 2008 election. Linzer (2013) built on this and combined information from historical forecasting models with results from public opinion surveys. Lauderdale and Linzer (2015) further fit the model to presidential election data from 1952 to 2012. Pickup and Johnston (2007) opened the door for space state models in Canada.

In the United Kingdom, Fisher, Ford, Jennings, Pickup, and Wlezien (2011) used a three-stage state-space model to forecast parliamentary seats from public opinion polls, by (1) adjusting and aggregating published vote-intention figures from different surveys, (2) forecasting how public

opinion could potentially change before election day and (3) predicting the seat totals from the forecasted election-day vote shares. Hanretty, Lauderdale, and Vivyan (2016) propositioned a similar model. Whiteley, Clarke, Sanders, and Stewart (2016) proposed a formal model based on the “Law of Cubic Proportions” which utilized the seats won by the major parties in the previous general election, together with vote intentions six months prior to the election, to forecast the distribution of seats in the House of Commons.

Likeminded models have been developed for a series of other European democracies. For example, Stoetzer, Neunhoeffer, Gschwend, Munzert, and Sternberg (2019) assessed the point predictions of different DLM specifications in the run-up to the German Federal election of 2013. By comparing polls, they found that only within the last month prior to election day, polls offered reliable information about the expected election outcome. Similarly, Walther (2015) tested the Jackman (2005) model in Germany and Sweden and showed that reasonable forecasts could be made despite the complexity of the systems and the emergence of new parties. Walther (2015) also introduced a “seasonal component” to account for cyclical changes in partisan support.

In comparison to the established democracies of North America and Europe, there is a striking lack of research on poll tracking and electoral forecasting in new democracies. Paradoxically, however, it is the latter that needs more attention than the former. Least developed democracies do not only have less informed electorates (see Fornos, Power, & Garand, 2004: 912), but also tend to report lower levels of post-election satisfaction than their counterparts (see Chu, Bratton, Lagos, Shastri, & Tessler, 2008). Unfortunately, few attempts have been made to advance in this direction.¹ As of this time, there are only two case studies that have used DLMs to forecast presidential elections in new democracies (see Bunker & Bau-chowitz, 2016; Cantú, Hoyo, & Morales, 2015).

In an effort to bridge this gap, the remainder of this article uses the evidence accumulated in established democracies to propose an approach based on poll aggregation and DLMs to track public opinion and forecast elections in new democracies. Section 4 begins by presenting an original two-stage model through which estimates can be computed and finishes by proposing a set of 11 countries in Latin America in which its accuracy can be tested. Section 5 begins by showing how the model fairs when it is applied to 26 elections (and 126 presidential candidates), and finishes by suggesting a combination of model specifications and structural characteristics that may explain why and when it is accurate.

¹ One effort to bridge this gap was made by Lewis-Beck and Bélanger (2012), in the special section Election Forecasting in Neglected Democracies published in this journal. However, the majority of the studies in the volume focused either on established democracies (Belgium, Japan, Norway and Spain), or on new democracies, but with a local election scope (Lithuania and Turkey) or a traditional forecasting method approach (Brazil).

4. The two-stage model

The two-stage model (TSM) follows Jackman (2005) and Walther (2015) in that it uses two sequential phases to process polling data. The purpose of the first stage is to input data and define weights (see Fisher et al., 2011; Pickup & Johnston, 2007). Its specific function is to weigh polls according to three criteria (described below in steps): accuracy track record, random error, and distance from the election. In turn, the purpose of the second stage is to output data and electoral estimates (see Jackman, 2005; Linzer, 2013). Its particular objective is to sequentially take the weighted data through a Bayesian inference process, in which posterior likelihood distributions can be simulated through a Markov chain Monte Carlo (MCMC) algorithm.

4.1. First stage: Poll calibration

The first step of the first stage is to account for the accuracy track record of each of the poll. The logic is that polls that are relatively more accurate in one election will be relatively more accurate in the next. To account for this logic, a weight is assigned to each poll based on the accuracy of its pollster in the previous election, by taking its final (or only) poll fielded in that cycle. Polls from a pollster with a more accurate poll in the previous electoral cycle are assigned a higher relative weight. Eq. (1) shows that the accuracy track record is computed by taking the sum of the absolute difference between the prediction ($\hat{\pi}$) for a candidate (i) and the actual result (π) for that same candidate, and dividing it by the total number of candidates (n)²:

$$\frac{1}{n} \sum_{i=1}^n |\hat{\pi}_i - \pi_i| \quad (1)$$

The second step of the first stage is to account for sources of random error that may affect the predictive capacity of each poll. The logic is that polls that anticipate lower levels of random error will be more accurate in comparison to those that anticipate higher levels of random error. To account for this logic, a weight is assigned to each poll based on its expected margin of random error; independent of the pollster. Polls that anticipate lower margins of random error are assigned a higher relative weight. Eq. (2) shows that the random error associated with each poll is computed by dividing a predetermined level of confidence (k) by the square root of the total number of persons interviewed (N) by the given poll (i)³:

$$\frac{k}{\sqrt{N_i}} \quad (2)$$

² To fit the model, track records are computed for every election in the sample. If a pollster fields a poll in election t , but did not field a poll in election in $t - 1$, it is given a rating equal to the worst pollster in election t .

³ To fit the model, the level of confidence was set to 95% ($k = 1.96$). Alternative values were tested (e.g., $k = 1.96$, $k = 1.65$, $k = 2.68$), yet results did not deviate significantly.

The final step is to account for the effect of time on each poll. The logic is that polls that are fielded closer to the election will be more accurate in comparison to those that are fielded further away. To account for this logic, a weight is assigned to each poll based on the number of days between the day the poll is published and election day; as above, also independent of the pollster. Polls fielded closer to the election are assigned a higher relative weight. Eq. (3) shows that the decay rate associated with each poll is computed by taking the exponential ($\exp$) of the multiplication between a fixed constant (λ) and the number of days between the day that the given poll is published and the day of the election (δ)⁴:

$$1 * \exp^{(\lambda * \delta)} \quad (3)$$

Once these computations are finalized, data are normalized within-steps and across-steps. First, they are normalized within-steps. This is done by taking the value computed for each observation (poll) and dividing it by the sum of all of the observations. This assigns each observation an individual bounded proportion ($[0,1]$). Then, data are normalized across-steps. This is done by simply adding up the three within-step weights and dividing it by three. This assigns each observation an aggregate unbounded weight ($[0, \infty]$). This final step is especially important not only because the within-step weights are measured on different scales, but also because it gives more accurate polls a disproportionately higher weight.⁵

4.2. Second stage: Bayesian updating

The second stage starts with the specification of a posterior model (see Berger & Wolpert, 1988; Bernardo & Smith, 2009). The posterior model (in this case, the forecast for each candidate) describes the probability distribution of all model parameters conditional on the observed data and some prior knowledge. It has two components: a likelihood, which includes information about model parameters based on the observed data, and a prior, which includes information collected before actually observing the data. Once these parameters are defined, and the data are collected, the likelihood and prior are simply combined using the traditional Bayes rule:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior} \quad (4)$$

To equate this specification to the matter in this study, consider a vector of polling data y , which is assumed to be a sample from a probability model with an unknown parameter vector θ . The objective, then, is to infer properties of θ based on the data y . The model is first represented by a likelihood function where $f(y_i|\theta)$ represents the probability density function of y_i given θ . It is

⁴ To fit the model, the constant was set to $-.08$ ($\lambda = -.08$). Alternative values were tested (e.g., $\lambda = -.07$, $\lambda = -.09$), yet results did not deviate significantly. The estimator is justified in Bunker and Bauchowitz (2016).

⁵ To fit the model, each step was weighed equally (1/3). An alternative distribution was tested (with 3/5 for accuracy track record, and 1/5 for each random error and decay rate), yet results did not deviate significantly.

then represented by a prior distribution in which θ has a probability distribution $p(\theta)$. And because both y and θ are random, Bayes theorem can be applied to derive the posterior distribution of θ given data y . Eq. (5) shows that $p(\theta|y)$ is given by $p(y|\theta)p(\theta)$ multiplied by $p(\theta)$ and divided by the marginal distribution of y :

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} \quad (5)$$

Because posterior distributions are rarely available in analytical forms and often involve multidimensional integrals, they can be estimated via simulation (Von Neumann, 1951). The most common way to conduct such simulations is through MCMC sampling methods (Tanner & Wong, 1987). While the term “Monte Carlo” refers to methods that rely on the generation of pseudorandom numbers, “Markov chain” refers to a sequence where each number is dependent on the previous one. Eq. (6) shows that θ at time t is given by a normal distribution of mean μ and variance σ , but is constantly updated until it converges in a posterior distribution (see Petris, Petrone, & Campagnoli, 2009):

$$\theta_t = N(\mu, \sigma) \rightarrow \theta_t = N(\theta_{t+1}, \sigma) \quad (6)$$

To update θ , and move through the chain, the TSM proposes a four-step adaptive Metropolis–Hastings (MH) algorithm (see Metropolis, Rosenbluth, Rosenbluth, Teller, & Teller, 1953; Metropolis & Ulam, 1949). The objective is to decide if the production of new values of θ are accepted or rejected (see Giordani & Kohn, 2010; Haario, Saksman, & Tamminen, 2001; Roberts & Rosenthal, 2009). To briefly explain the process, let $q(\cdot)$ be a probability distribution and θ_z the starting state. Then, at each step t , a proposal state θ_z is generated conditional on the current state. After drawing uniform random numbers, θ_z is accepted or rejected, and updated, according to the previously defined acceptance probability. This is iterated for $t = 1, \dots, T-1$.

4.3. Generating a forecast

To clarify how the two stages come together, consider a contest in which a variable number of polls (p) publishes the voting intention (v) for a variable number of candidates (k) in the days leading up to an election. In the first stage, each k is assigned a weight. In the second stage, the weighted data (pvk) is entered all at once into the Bayesian model. The combination of the likelihood and the prior generates a probable result for each candidate. Then, the probable result is entered sequentially into the MCMC algorithm. The result of the iterative simulation process generates a definitive forecast for each candidate. In this study, the definitive forecast is the product of all polls available.⁶

⁶ An individual forecast can be produced at any moment during a campaign cycle, considering there are at least two polls which can be assigned a weight. A forecast can be generated at the earliest on the date of the second poll, and at the latest on the date of the last poll. Thus, a timeseries forecast can be produced by simply computing a forecast for each day in a given campaign cycle. Considering that for this study all of the data (polls) were available, the TSM forecast is set to be generated on the last possible date in the campaign cycle.

4.4. Fitting the model

To fit the TSM to new democracies, political and electoral particularities are considered. For example, their institutional procedures. While the majority of established democracies hold parliamentary elections, most new democracies hold presidential elections (Cheibub, 2007). The implication for the TSM model is important. Since the parameter of interest in the former is related to the sum of seats for each party, in the latter it is related to the share of votes for each candidate (either in the first or second round). Thus, the TSM uses survey data that measures voting intention for particular presidential candidates, rather than survey data that represent electoral popularity for individual parties or coalitions.

Another key difference between established and new democracies is related to electoral patterns. Previous research has found that in the latter voters are more volatile (Kreuzer & Pettai, 2003; Mozaffar & Scarritt, 2005; Roberts & Wibbels, 1999; Tavits, 2005) and split-ticket voting is common (Mainwaring & Scully, 1995; Shugart, 1995). The distinction is relevant. Since elections are often contested by different parties (and candidates) from one election to the next, estimates of their support are less consistent. Thus, instead of assuming a preexisting distribution, the TSM is fit with uninformative flat priors with an unknown variance. This is also known as the “let the data speak for itself” approach (Gelman, Carlin, Stern, Dunson, Vehtari, & Rubin, 2013: 51).

A third key difference between established and new democracies is related to the party landscape. Previous research shows that party systems tend to be less institutionalized (Mainwaring & Scully, 1995; Riedl, 2008; Stockton, 2001) and more fragmented (Gallagher & Mitchell, 2005) in the latter. The difference is important. Given that there are higher turnover rates in the set of dominant parties, electoral preferences are less stable over time. Thus, instead of using a high number of MCMC iterations, the TSM is fit with a number of simulations closer to the average lower bound, with 30,000 simulations with 5000 burn-ins, but high enough to satisfy the Gelman–Rubin diagnostic (see Brooks & Gelman, 1998; Gelman & Rubin, 1992).

5. Data

To test the model, data were collected from the new democracies of Latin America. As the third largest democratic region, after North America and Europe, it presents the perfect conditions for the experiment. The majority of countries use presidential elections to elect their leaders and present high rates of volatility and party system fragmentation. Furthermore, while there is some research on forecasting (see Traugott, 2015), it has been mainly limited to traditional methods (e.g., Beltrán & Valdivia, 1999; Turgeon & Rennó, 2012). And, while there are some studies regarding DLMS, as of this time, there are only two peer-reviewed articles that look at countries in the region (Bunker & Bauchowitz, 2016; Cantú et al., 2015).

Because the TSM uses only polls as input, three considerations were taken into account in order to ensure

Table 1

Summary of the data.

Source: Media outlets and pollsters.

Country	Year	Pollsters	Polls	Candidates	Valid preferences
Argentina	2003	8	8	6	89.4
	2007	10	19	4	72.7
	2011	12	26	6	89.5
	2015	19	85	3	80.7
Brazil	2006	5	33	4	82.3
	2010	4	44	3	82.6
	2014	4	15	3	82.6
	2018	16	24	8	80.7
Chile	1999	4	7	6	87.3
	2005	7	10	4	84.8
	2009	11	33	4	84.6
	2013	7	12	9	84.1
	2017	7	108	8	58.7
Colombia	2002	4	17	4	84.7
	2006	3	9	4	82.3
	2010	5	18	9	90.4
	2014	5	17	6	90.3
	2018	8	39	6	84.5
Costa Rica	2010	3	10	5	86.3
	2014	4	13	5	75.4
	2018	5	19	7	57.8
Ecuador	2009	4	18	3	73.5
	2013	8	31	5	79.2
	2017	9	25	4	69.3
El Salvador	2014	8	14	2	75.7
	2019	12	29	3	81.9
Mexico	2000	10	28	4	100.0
	2006	8	18	4	94.8
	2012	13	50	4	92.5
	2018	22	49	4	86.6
Paraguay	2013	7	13	4	82.9
	2018	5	7	2	88.3
Peru	2006	3	6	5	78.8
	2011	2	18	5	78.6
	2016	6	27	5	60.5
Uruguay	2009	4	14	4	91.4
	2014	7	28	4	90.0
TOTAL	–	279	941	176	–

Note: Table A1 elaborates on the definition and measurement of each variable; Table A2 shows the full list of pollsters that fielded at least one poll in each election.

transparency and guarantee replicability. First, only polls from pollsters that conducted their fieldwork before the election were registered. Second, only polls from pollsters that released their results into the public sphere were registered. Finally, only polls with information regarding the date of the fieldwork, the number of people interviewed, as well as information concerning the processing method, and the distribution of electoral preferences for all major candidates in the race were registered.⁷ Table 1 shows the summary of the total amount of data used in this study.

It is important to note that in the dataset each row corresponds to an individual poll with a vector (v) of information on the vote intention for each of the candidates. In the best-case scenario $v = 100$. This means that the

sum of valid preferences is equal to 100%. However, it is often the case that $v < 100$. This occurs either because (1) a poll is fielded early in the electoral cycle and does not capture information for candidates that register later in the race, or because (2) a poll does not report voting intention for all of the candidates. When this occurs, the vector is simply rescaled to reflect valid preferences only (ergo, $v < 100 \rightarrow v = 100$). At any rate, to fit the TSM, only polls fielded in the three months (90 days) prior to the election were considered.

6. Results

How do voters inform themselves during campaign cycles and anticipate the results of elections? The above-reviewed literature suggests that they may use any number of methods to do so. However, the predominant manner is through polls (Lau & Redlawsk, 2006; West, 1991). Thus, in what follows, the TSM forecast is

⁷ Vote intention is understood as the answer to the following question: "If the election took place [today/this week], who would you vote for?" Since the wording slightly differed across pollsters, this study only considered polls in which it was clear that the question was framed as a voting intention question.

compared to the average poll prediction (see Campbell, 2012).⁸ The premise that grounds the comparison is that in the final stretch of an electoral cycle, voters are normally exposed to an overwhelming number of polls that are released into the public sphere. Since they cannot know which of all the polls will be the most accurate, they will tend to maximize estimations by implicitly averaging their results.⁹

Table 2 compares the mean absolute error of the TSM to the mean absolute error of the polls at the election level. It shows their difference and a column that indicates whether the model is more accurate than the polls. Remarkably, the model is indeed more accurate than the polls, producing a more precise estimation for every election in the sample—the model errs by an average of 4.1 points, whereas the polls err by an average of 5.2 points. While the overall error produced by both estimators is arguably high, the margin by which the model is superior makes a significant difference. For instance, in the 2012 Mexican presidential election the model outperformed the polls by an average of 3%.

Fig. 1 plots the election-level precision of the TSM against the election-level precision of the polls. The 45-degree dashed line ($x = y$) represents the perfect equality between both variables. When the TSM is more accurate, the observations (elections) fall above the line; and when the polls are more accurate, they fall below the line. Remarkably, the plot shows that all of the observations fall above the line. The distance of each election from the line illustrates the magnitude of the accuracy of the TSM. While the pattern shows that the TSM is especially efficient in some elections where overall error is low (such as Mexico 2012), it also shows that it is efficient in elections where overall error is high (such as Peru, 2011).

To supplement the analysis, the election-level data are disaggregated into candidate-level data (see Table A3 for the full list). Remarkably, the model is once again more accurate than the polls; making a relatively more precise forecast for 95% of the candidates (121 of 128). Results show that when the model is more accurate, it outperforms the polls by an average of 1.2%. In contrast, when the polls are more accurate, they outperform the model by only 0.3%. While it may not seem like a significant margin, it is large enough to decide elections. For example, in the 2018 election of Mexico, the model forecasted that Andrés Manuel López Obrador would obtain 50.1% of the vote,

⁸ This is the standard parameter for comparison in the literature. Most of the studies that compare forecasts to polls use the mean absolute error (MAE) of polls during the campaign cycle as the benchmark (see Armstrong et al., 2015; Erikson & Wlezien, 2008; Graefe et al., 2014; Jennings & Wlezien, 2018).

⁹ There is vast evidence that the majority of voters make up their mind even before the campaign starts (Dalton, McAllister, & Wattemberg, 2000; Gopoian & Hadjiharalambous, 1994; Lau & Redlawsk, 2006). Thus, there are compelling arguments to use the average poll prediction (or MAE) during the full campaign cycle (approximately 90 days) as the benchmark. At any rate, as a robustness check, the average poll prediction using the three polls published in each election were computed as an alternative standard, yet results did not deviate significantly from those presented below. Figure A1 and Figure A2 show the results.

Table 2
Election-Level errors.
Source: Author.

Country	Year	TSM error	Poll error	Dif.	Better?
Argentina	2007	6.15	6.88	0.73	Yes
	2011	2.10	3.70	1.61	Yes
	2015	4.40	4.54	0.14	Yes
Brazil	2010	6.89	7.97	1.08	Yes
	2014	8.92	9.48	0.56	Yes
	2018	2.66	3.20	0.54	Yes
Chile	2005	1.63	2.38	0.75	Yes
	2009	0.84	2.18	1.34	Yes
	2013	1.41	2.54	1.13	Yes
	2017	2.09	2.95	0.86	Yes
Colombia	2006	3.16	4.26	1.10	Yes
	2010	4.67	4.90	0.23	Yes
	2014	5.39	6.59	1.19	Yes
	2018	3.22	4.54	1.32	Yes
Costa Rica	2014	8.02	8.78	0.76	Yes
	2018	5.96	7.43	1.48	Yes
Ecuador	2013	3.26	4.41	1.15	Yes
	2017	4.26	5.51	1.25	Yes
El Salvador	2019	2.80	3.52	0.72	Yes
Mexico	2006	1.45	2.65	1.21	Yes
	2012	1.57	4.85	3.28	Yes
	2018	3.17	5.07	1.91	Yes
Paraguay	2018	5.27	6.54	1.27	Yes
Peru	2011	8.85	10.26	1.42	Yes
	2016	5.82	7.40	1.59	Yes
Uruguay	2014	1.92	2.09	0.18	Yes

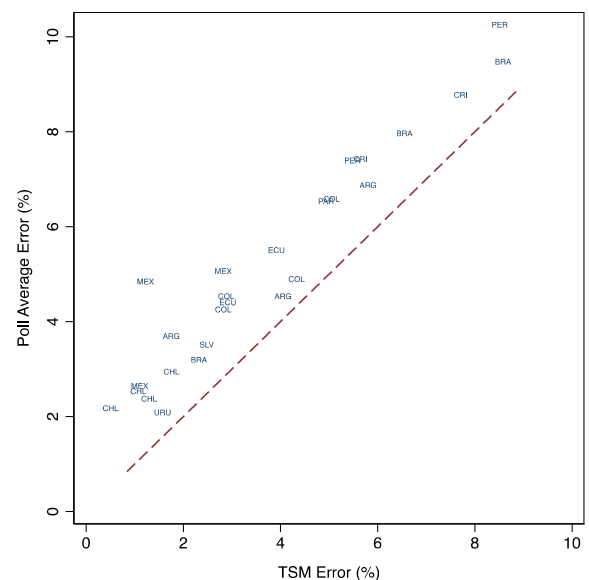


Fig. 1. Election-level precision.
Source: Author.

while the polls predicted 48.7%. Calderón won the election with 53.2%.

Fig. 2 plots the candidate-level precision of the TSM against the candidate-level precision of the polls. As above, the dashed line represents the 45-degree equilibrium.

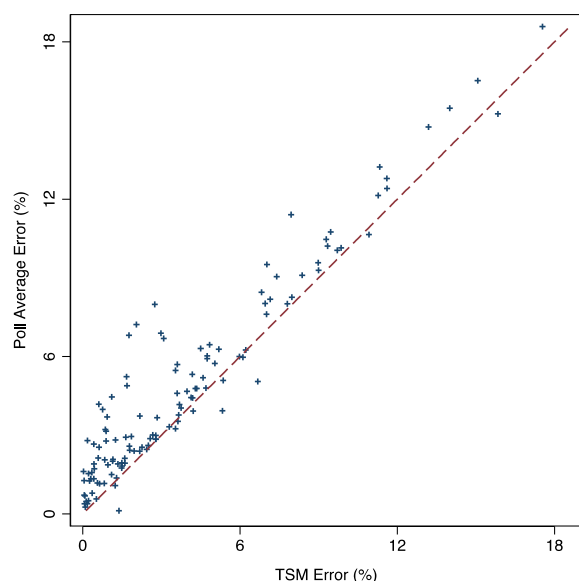


Fig. 2. Candidate-level precision.
Source: Author.

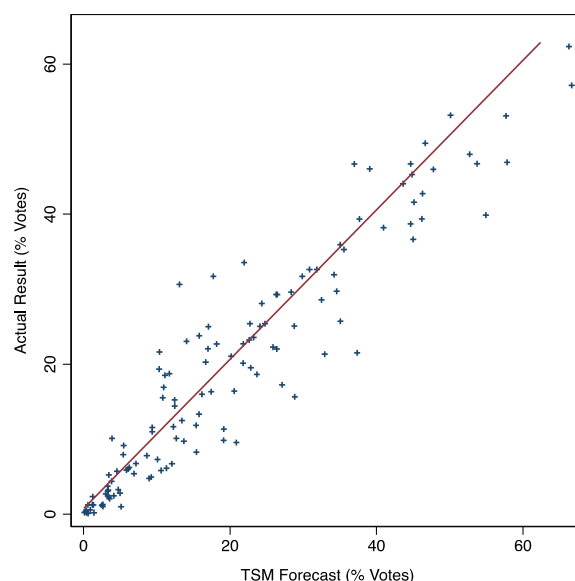


Fig. 3. Candidate-level fit.
Source: Author.

When the TSM is more accurate, the observations (candidates) fall above the line; and when the polls are more accurate, they fall below the line. Once again, the pattern shows that the TSM is substantially superior to the polls. The great majority of the observations fall above the line. The average poll prediction was only superior to the TSM forecast for six candidates, by an average advantage of only 0.26%.

Fig. 3 plots the linear fit between the TSM forecast and the actual result for each candidate. As expected, it shows that the majority of the observations are on (or around) the linear-fit; the r -squared value is both high (0.89) and significant ($p < 0.000$). Accuracy levels, however, are not consistent across the board. While the distribution around the linear fit converges for candidates with high support ($< 35\%$) and low support ($> 15\%$), it deviates for candidates in the “middle ground” ($15\% > 35\%$). Thus, while the TSM tends to produce relatively accurate estimates for both frontrunners and tailgaters, it diverges for candidates who are closer to the average electoral result.

To further examine this issue, Fig. 4 plots the linear fit for the TSM forecast and the actual result for each candidate, controlling for election placement. The results confirm that deviations are higher for those in the “middle ground”. While first- and fourth-place candidates are explained by around 70% of the variability of the forecast, second- and third-place candidates are predicted by around 50%. At any rate, the model correctly forecasted placement for 86 of the 126 candidates in the sample, including an overwhelming majority of those who placed first. And while the model incorrectly forecasted placement for the rest of the candidates, in the majority of the cases it was only off by a few decimals, and rarely by more than one place.

The placement issue can be illustrated by using the 2017 presidential election of Chile as an example. While

the forecast correctly placed six of the eight candidates in competition, it misplaced the fourth- and fifth-place candidates. Indeed, the model forecasted Carolina Goic to come in fourth and José Antonio Kast to come in fifth, when in reality the reverse occurred: Kast placed fourth and Goic placed fifth. Albeit, while the model erred in the placement of the candidates, it nonetheless accurately forecasted the result for each of them (with estimates within credibility intervals). Meanwhile, the polls did not only err in the placement of both of the candidates, but also incorrectly predicted their vote shares (with estimates outside margins of error).

At any rate, the model correctly forecasted placement for all of the candidates in the majority of the elections, including Argentina 2007, 2015; Brazil 2010; Chile 2005, 2009; Colombia 2006, 2018; Ecuador 2007; Mexico 2012, 2018; Paraguay 2018; Peru 2016; El Salvador 2019; and Uruguay 2014. This is remarkable considering that in most of these elections polls reported conflicting results. For example, in the 2016 presidential election of Peru, all of the polls leading up to election night predicted a technical tie for the second place, between Pedro Pablo Kuczynski and Verónica Mendoza. Finally, Kuczynski came in second, advanced to the runoff round, and won the presidency (as the TSM forecasted).

Why is the forecast more accurate in some elections? The answer may lie in the particular model specifications. Because it only uses polls as input, the level of accuracy may be determined by the number of pollsters or polls taken into account. Or, because polling data are naturally incomplete (recall vector v), the level of accuracy of the model may be determined by the number of candidates or the sum of valid preferences. Alternatively, the answer may lie in structural variables that are common to most new democracies in Latin America. In this latter scenario,

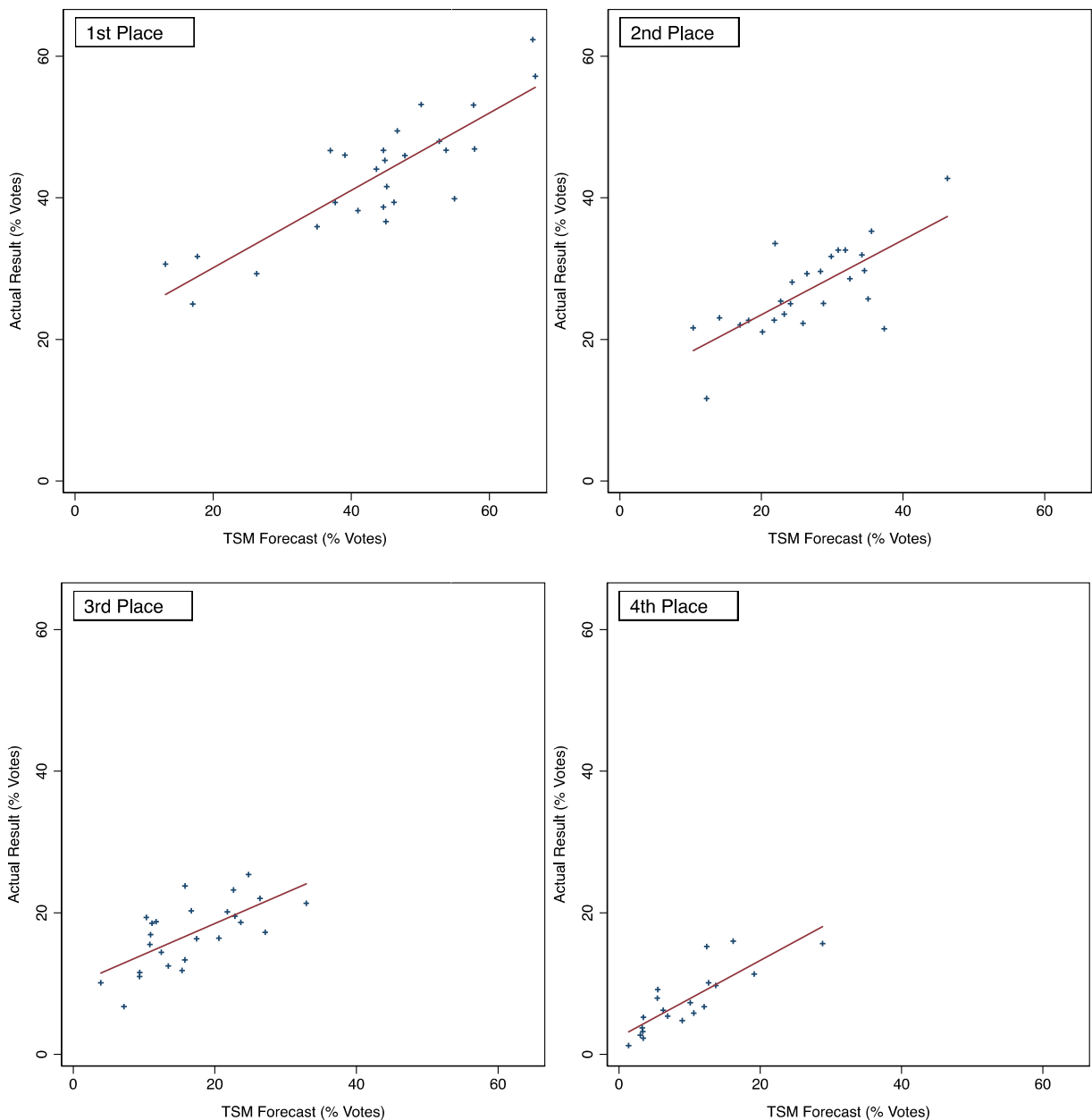


Fig. 4. Candidate-level fit, by election placement.
Source: Author.

the level of accuracy of the TSM may instead be determined by factors such as institutional procedures, or those related to electoral patterns or the party landscape.

Table 3 shows a linear regression that seeks to explain election-level errors considering model specifications and structural variables. Model specifications variables are described above. Structural variables are operationalized as follows. Institutional procedures are framed as presidential election rules, and are measured on

a permissive-restrictive scale (1=simple majority, 2=simple majority with reduced threshold, 3=absolute majority); electoral patterns are measured using Cohen, Salles Kobilanski, and Zechmeister's (2018) party stability volatility index (higher numbers indicate more volatility); and party landscape is measured using Laakso and Taagepera's (1979) party fragmentation index (higher numbers indicate more electoral parties).

To run the regression, model specifications were entered sequentially while structural variables were held

Table 3
Explaining election-level errors.

	M1	M2	M3	M4	M5
Model specifications					
Pollsters	−0.330*** (0.10)				−0.424*** (0.130)
Polls		−0.077*** (0.040)			0.062 (0.057)
Candidates			−0.333 (0.281)		−0.381* (0.213)
Valid preferences				−0.072 (0.044)	−0.026 (0.046)
Structural variables					
Presidential election rules	−1.981*** (0.466)	−1.647** (0.686)	−1.383 (0.807)	−1.068 (0.802)	−1.889*** (0.527)
Party fragmentation	0.260** (0.121)	0.281 (0.181)	0.301* (0.168)	0.177 (0.1883)	0.285** (0.108)
Electoral volatility	0.314 (1.284)	−0.253 (1.441)	0.594 (1.426)	−0.080 (1.403)	1.162 (1.107)
Constant	9.383*** (1.247)	7.665*** (1.768)	6.783*** (1.889)	11.580*** (3.741)	12.171*** (3.530)
Regression fit					
N	26	26	26	26	26
Root MSE	1.894	2.236	2.230	2.258	1.842
R-squared	0.463	0.251	0.209	0.236	0.564

Dependent variable: Forecast absolute mean error. Note: Robust Standard Errors in parentheses.

***Sig.: $p < 0.001$.

**Sig.: $p < 0.05$.

*Sig.: $p < 0.01$.

constant. The results suggest that the number of pollsters included in the dataset have the strongest effect on error. For every additional pollster considered for an individual election, the TSM's mean error fell by around 0.42% (see M5). The results also suggest that elections that are contested under more restrictive rules (when the president is elected by simple majority) produce lower errors than election contested under more permissive ones (when the president is elected by absolute majority). At any rate, these finding should be interpreted cautiously, considering the low sample size.

7. Discussion

The results show that the TSM is effective in reducing the difference between poll predictions and results. This is exceptionally valuable considering that all of the forecasts were produced using data only available prior to elections. In other words, forecasts were simulated without considering any post-election information. If a voter would have had no knowledge of the “true” state of an election, they would have been more accurately informed by the TSM forecast than by the average poll. This is especially relevant in the context of new democracies, and especially of Latin America, where accuracy levels of polls have been harshly scrutinized in recent years (see [Serrano & Navarro, 2017](#)).

While the results are encouraging, they do not mean that the model is watertight. While the TSM does indeed produce more accurate forecasts at the election and candidate levels, it does not always produce overall accurate

estimates. The results suggest that deviations may be due to the number of pollsters included in the sample: elections in which more pollsters fielded polls show lower levels of error. Naturally, this is problematic. Not so much because of the availability of data, but because in some elections only a handful of pollsters' field polls. If in any given election only one pollster fields polls, it is likely that error levels will be high. However, it is reassuring to know that this is rarely the case (see Table A2); the polling industry in most new democracies is blooming.

8. Conclusion

This article has proposed a method to minimize the difference between electoral predictions and electoral results. It has built on findings that stem from established democracies, where most of the research has been carried out, but has focused on filling the gap for new democracies, which have been thus far largely neglected by the literature. It has propositioned a TSM and computed estimates with Bayesian algorithms and Markov chains. It has tested the model using data stemming from 11 countries and 26 elections in Latin America and produced remarkably precise results. In comparison to polls, not only did the model output a more accurate forecast for every election in the sample, but also it did for nine out of every ten candidates.

Its main contribution has been to extend a line of literature that has almost exclusively developed in established democracies. Not only are there few studies that have generally looked at forecasting in new democracies,

but there are only two that have particularly used state space representation DLMS, and both of them are case studies (see Bunker & Bauchowitz, 2016; Cantú et al., 2015). As such, the article not only opens up the field to compare DLMS across types of democracy, but also provides a benchmark for future research in developing nations. In particular, it provides a reference for the study of public opinion and elections in Latin America.

A secondary contribution is to have identified avenues for future research. One of them is analytical in nature and suggests that there is much room to further explain why the TSM's errors are related to some elections and not to others. One open question is whether the model more accurate than other forecasting methods (such as those produced by quantitative models or betting markets), or it is only superior to polls? Furthermore, there is also space to expand the TSM to other new democracies. For instance, does the TSM also produce accurate results in post-communist European countries that have been found to respond to different dynamics, as Stegmaier and Lewis-Beck (2009) have suggested?

A second avenue of research is experimental in nature and suggests that there is much space to further improve the TSM's accuracy levels. For example, does the model's general specification and particular settings (developed to fit the particular context of the new democracies in Latin America) provide an optimum solution to the “accuracy problem”, or can they be further developed and tailored to reduce errors? Also, is the parsimonious and minimalist polls-only approach the most efficient manner of modeling electoral outcomes, or could the inclusion of more complex information (such as social media data) help increase the accuracy level of the model, as Armstrong et al. (2015) have suggested?

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ijforecast.2020.02.004>.

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